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Phase-field model of dendritic growth

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Abstract

The phase-field models for binary and ternary alloys are introduced, and the governing equations and phase-field parameters for dilute alloys are derived at a thin interface limit. The phase-field simulations on isothermal dendrite growth for Fe-C, Fe-P and Fe-C–P alloys are carried out and the effect of the ternary alloying element on dendrite growth is examined. The secondary arm spacing for Fe-C, Fe-P and Al–Cu alloys is numerically predicted using the phase-field model and compared to the experimental data. The change in the arm spacing, and the exponent of local solidification time depending on alloy is systematically examined by imposing artificial set of physical properties. The phase-field simulation for the microstructure evolution during rapid solidification is also successfully carried out. Through the numerical examples, the wide potentiality of the phase-field model to the applications on solidification has been demonstrated. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

The phase-field model is known to be a powerful tool for describing the complex pattern evolution of the interface between mother and new phases in nonequilibrium state because all the governing equations are written in unified manner in the whole space of system. The model was originally proposed for simulating dendrite growth in undercooled pure melts [1–9] and has been extended to solidification of alloys [10–20]. The first model for alloy solidification was proposed by Wheeler et al. (WBM model) [10,13]. The WBM model has been the most widely used [10–13,21,22] and is derived in a thermodynamically consistent way [13]. In the model, any point within the interfacial region is assumed to be a mixture of solid and liquid both with a same composition. The phase-field parameters are determined not only under a sharp interface condition [10], but also under a finite interface thickness condition [23].

Recently, Kim et al. have proposed a new phase-field model for binary alloys (KKS model) [24]. The model is equivalent with the WBM model, but has a different definition of the free energy density for the interfacial region. An extra-potential originated from the free energy density definition in the WBM model disappears and the governing equations completely correspond to those for pure

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materials [25] as pointed out by Langer [26]. In the present study, the phase-field model for binary alloys by Kim et al. and its extension to ternary alloys are briefly introduced. The examples of numerical analyses for different types of dendrite growth are demonstrated and the validity of the models is discussed.

2. Governing equations

2.1. Free energy density

In the KKS model, the free energy density, $f(c,\phi)$, where ϕ is phase-field, is defined as the sum of the free energies of liquid and solid phases with different compositions of c_L and c_S, respectively, and an imposed double-well potential, $Wg(\phi)$. The solute composition in interface region, c, is determined to be the fraction-weighted value of both liquid and solid compositions. The chemical potentials, μ, defined as the difference between the chemical potentials of solute and solvent, are selected as a thermodynamic variable and those of solid and liquid phases are assumed to be equal at any point within the interface region:

$$f(c,\phi) = h(\phi)f^{S}(c_S) + (1-h(\phi))f^{L}(c_L) + Wg(\phi), \tag{1a}$$

$$c = h(\phi)c_S + (1-h(\phi))c_L, \tag{1b}$$

$$\mu^{S}(c_S(x,t)) = \mu^{L}(c_L(x,t)), \tag{1c}$$

where $h(\phi) = \phi^2(3-2\phi)$, $g(\phi) = \phi^2(1-\phi)^2$, the subscripts S and L indicate solid and liquid phases, respectively. Note that the chemical potentials, μ^S and μ^L, are derived from the free energy densities, f^S and f^L.

The advantage of the above phase-field model lies in the definition of the free energy density functional in the interface region at an equilibrium state. The free energy density at the interface without the $Wg(\phi)$ are defined as the fraction-weighted average values of the free energy in solid and liquid phases and is given by $h(\phi)f^S(c_S) + (1-h(\phi))f^L(c_L)$. Since the equilibrium compositions of solid and liquid phases are determined to keep each chemical potential equal, this equation corresponds to the common tangent line in the free energy–composition diagram itself.

2.2. Governing equations using dilute solution approximation

A phase-field model for solidification is based on the Ginzburg–Landau free energy functional. The phase-field and diffusion equations are obtained in a thermodynamically consistent way [24]:

$$\frac{\partial\phi}{\partial t} = M[\varepsilon^2\nabla^2\phi - h'(\phi)\{f^S - f^L\} - Wg'(\phi)], \tag{2}$$

$$\frac{\partial c}{\partial t} = \nabla\left(\frac{D(\phi)}{f_{cc}}\nabla f_c\right), \tag{3}$$

where M and ε are phase-field parameters, $D(\phi)$ is solute diffusion coefficient and subscripts under f mean the first or second derivatives by corresponding variables.

In Eqs. (2) and (3), the explicit forms of the terms are dependent on the definition of the free energy density, f_ϕ, f_c and f_{cc}, for the interface region. With the free energy density for a dilute alloy, f_ϕ, f_c and f_{cc} in Eqs. (1a)–(1c) are expressed as functions of c_S, c_L and ϕ. The resulting phase-field and diffusion equations turn out to be

$$\frac{1}{M}\frac{\partial\phi}{\partial t} = \varepsilon^2\nabla^2\phi - Wg'(\phi) + \frac{RT}{V_m}h'(\phi)\ln\frac{(1-c_S^e)(1-c_L)}{(1-c_L^e)(1-c_S)}, \tag{4}$$

$$\frac{\partial c}{\partial t} = \nabla\left[D(\phi)\{(1-h(\phi))c_L(1-c_L). + h(\phi)c_S(1-c_S)\}\nabla\ln\frac{c_L}{1-c_L}\right], \tag{5}$$

where R is gas constant, T is temperature, and V_m is molar volume.

The phase-field parameters of ε and W are related to interface energy, σ, and interface width, 2λ, where ϕ changes from 0.1 to 0.9, and the parameter, M, is related to the kinetic coefficient, β [23,24]. They are obtained also with a dilute solution approximation and are given by

$$\sigma = \frac{\varepsilon\sqrt{W}}{3\sqrt{2}}, \tag{6}$$

$$2\lambda = 2.2\sqrt{2}\frac{\varepsilon}{\sqrt{W}}, \tag{7}$$

$$\beta = \frac{V_m}{RT}\frac{m^e}{1-k^e}\left(\frac{\sigma}{M\varepsilon^2} - \frac{\varepsilon}{D_i\sqrt{2W}}\varsigma(c_S^e, c_L^e)\right), \tag{8a}$$

$$\varsigma(c_S^e, c_L^e) = \frac{RT}{V_m}(c_L^e - c_S^e)^2 \int_0^1 \frac{h(\phi)[1-h(\phi)]}{[1-h(\phi)]c_L^S(1-c_L^e) + h(\phi)c_S^L(1-c_S^e)} \frac{d\phi}{\phi(1-\phi)}, \tag{8b}$$

where m^e is equilibrium slope of liquidus, k^e is equilibrium partition coefficient and $D_i$is diffusion coefficient in interface region. Since the interface temperature, T^i, is related to interface velocity, V, and solid interface composition, c_S^i, by

$$T^i = T_M - m^e\frac{c_S^i}{k^e} - \beta V, \tag{9}$$

where T_M is melting temperature and the kinetic coefficient, β, is defined to be the inverse of the usual linear kinetic coefficient, μ_K. The above equation for β permits vanishing kinetic coefficient or an infinite value of μ_K by adjusting the parameters to cancel out two terms from each, as in the thin-interface phase-field model for pure materials [25].

In a ternary alloy, the free energy of solid and liquid phases are written as

$$f^S = \sum_{i=1}^{3} c_i^S\mu_i^S(T), \quad f^L = \sum_{i=1}^{3} c_i^L\mu_i^L(T) \quad (\mu_i^S = \mu_i^L \text{ within the interface}), \tag{10a}$$

$$\sum_{i=1}^{3} c_i = 1. \tag{10b}$$

With the free energy density given above, the phase-field equation and diffusion equations can be easily derived as the similar ones as Eqs. (4) and (5). Note that the cross terms in the diffusion equations are still under discussion and only a dilute ternary alloy model is proposed [28]. The phase-field mobility in the thin interface limit is also available with the vanishing kinetic coefficient:

$$\frac{1}{M} = \frac{\varepsilon^3}{\sigma\sqrt{2W}}\left(\frac{1}{D_{1i}}\zeta_1(c_{1L}^e, c_{1S}^e) + \frac{1}{D_{2i}}\zeta_2(c_{2L}^e, c_{2S}^e)\right). \tag{11}$$

3. Calculations

For the simulation, the phase-field equation was modified so as to include the anisotropy of interface. The four-hold anisotropy was introduced by putting the coefficient in the phase-field parameter as follows:

$$\varepsilon(\theta) = \varepsilon\{1 + v\cos(4\theta)\}, \tag{12}$$

where v was the magnitude of anisotropy ($v = 0.03$) and θ is the angle between normal direction of interface and x-axis. Note that the ε^2 should be located inside the divergence in Eq. (4) and the following terms should be added when the anisotropy is considered [15]:

$$\frac{\partial}{\partial x}\left[\varepsilon(\theta)\varepsilon'(\theta)\frac{\partial\phi}{\partial y}\right] - \frac{\partial}{\partial y}\left[\varepsilon(\theta)\varepsilon'(\theta)\frac{\partial\phi}{\partial x}\right].$$

In the calculation, Eqs. (1b), (1c), (4) and (5) were used. The equations were discretized on uniform grids using an explicit finite difference scheme. During the calculation, a stochastic noise on the liquid composition in the vicinity of interface was imposed in order to simulate the fluctuations, which gave the well-developed secondary arms. The physical properties of alloys used in calculation are shown in Table 1. Note that the free energy densities are obtained using the data for binary alloys under the assumption of a dilute solution approximation. The phase-field simulations were carried out for isothermal dendrite growth, growth of dendrite secondary arms and the microstructure evolution during rapid solidification, and the calculation condition for each was described in the following sections.

3.1. Isothermal dendrite growth

The isothermal dendrite growth of Fe–0.5 mol%C binary alloy and Fe–0.5 mol%C–0.001–0.02 mol%P ternary alloys was simulated at different temperatures. The calculation area of

Table 1
Physical properties of alloys

Properties	Fe–C	Fe-P	Al–Cu
σ (J/m^2)	0.204	0.204	0.093
T_m (K)	1810	1810	933.3
D_L (m^2/s)	2.0×10^{-8}	1.7×10^{-9}	3.0×10^{-9}
D_S (m^2/s)	6.0×10^{-9}	5.5×10^{-11}	3.0×10^{-13}
V_m (m^3/mol)	7.7×10^{-6}	7.7×10^{-6}	10.5×10^{-6}
k_e	0.204	0.102	0.14
m_e (K/mol)	−1836	1802	−620

500×1000 meshes was prepared and a small triangle solid of 20×20 meshes was initially put in the corner of the area. Mesh size was 1.0×10^{-8} m^2 and the system temperature was varied from 1765 to 1785 K.

3.2. Growth of dendrite secondary arms

The Fe–0.5 mol%C, Fe–0.61 mol%P, Fe–0.5 mol%C–0.02 mol%P and Al–1.96 mol%Cu alloys were selected as sample alloys. A two-dimensional rectangular calculation area was prepared. The width of the area (400–500 meshes) was set to about 10 times the length of the experimentally reported secondary arm spacing, and the height (125–400 meshes) was to be the same as the length of the experimental primary dendrite arm spacing. Small triangle solids were put in the bottom and their size and number were determined by the preliminary calculations so as to obtain the secondary arm spacing independent of the condition. The mesh size used in calculations was 0.5×10^{-6} m. The system temperature was reduced constantly from the liquidus temperature with a constant cooling rate from 100 to 5000 K/s.

3.3. Microstructure evolution during rapid solidification

For simulating the microstructure evolution during the initial stage of rapid solidification, a heat conduction equation was solved simultaneously by the phase-field and diffusion equations. Since the thermal diffusivity of the alloy was about hundred times larger than the diffusional one, the mesh size for thermal calculation was taken to be 10 times larger than that for phase-field calculation. The latent heat generation in a thermal calculation element was calculated from the change in the total phase-field values inside a thermal element. The Fe–0.5 mol%C alloy was selected in order to predict the microstructure of low carbon steel by strip casting process. Two-dimensional rectangular calculation areas with different sizes were prepared for calculations. The one was for the phase-field and solute field, and of 1000×2000 meshes. The other was for the thermal field and of 100×2000 meshes. The mesh sizes for phase-filed and thermal calculations were 0.5×10^{-7} and 0.5×10^{-6} m, respectively.

It was assumed that the system was contacted with a chill plate of constant temperature of 600 K and cooled by Newtonian cooling with the heat transfer coefficient of 10000 W/Km^2. The phase-field calculation has been started at a certain degree of undercooling, where a small triangle solid of 10×10 meshes was initially put in the corner of the area. The initial undercooling from the liquidus temperature of 1800 K was varied from 5 to 15 K.

4. Results and discussion

4.1. Isothermal dendrite growth

The dendrite shapes of Fe–0.5 mol%C binary alloy and Fe–0.5 mol%C–0.001 mol%P ternary alloy at 1780 K are shown in Fig. 1. The secondary arms develop well and the arm spacing becomes narrow with a small additional phosphorous because the phosphorous is likely to enrich at the

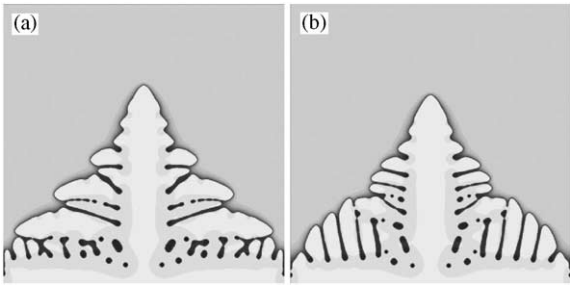


Fig. 1. The change in dendrite shape of steel due to ternary alloying element: (a) Fe–0.5 mol%C; (b) Fe–0.5 mol%C–0.001%P.

interface and reduces the interface stability. However, the dendrite shape becomes coarse with the further increase of phosphorus concentration. It is due to the decrease of the phase-field mobility because the value of ζ_2 in Eq. (11) decreases with increase of phosphorous concentration. The change in the tip velocity is roughly proportional to the phase-field mobility and the tip radius increases depending on the decreasing tip velocity. These results show that a small addition of phosphorus changes secondary arm spacing significantly even when it does not change the interface velocity much.

4.2. Growth of dendrite secondary arms

The growth and the selection of secondary dendrite arms for Fe–0.5 mol%C alloy are shown in Fig. 2. All the initial solid seeds grow but the shapes of the arms are slightly different from each other due to the imposed noise. The advanced arms grow preferably and other arms stop advancing or slightly melt back due to the curvature effect. After the selection of the secondary arms, the coarsening of the selected arms is observed. Kahlweit has suggested that theoretically the coarsening of secondary arms is due to melting back of the arms. Others have suggested that arm selecting mechanisms such as arm thinning or arm separation have not been observed in the phase-filed calculation. The arm selection in the phase-field model seems to be reasonable because only melting back process is observed in the experiment using a transparent organic material [29].

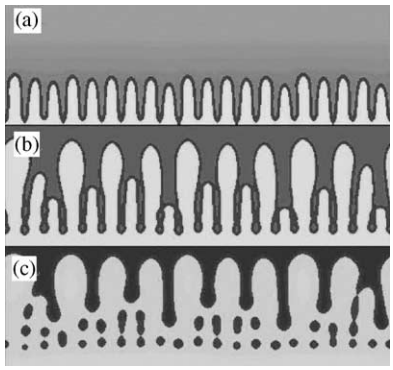


Fig. 2. Growth of secondary arms for Fe–0.5mol%C (5000 K/min): (a) 0.045, (b) 0.09 and (c) 0.15 s.

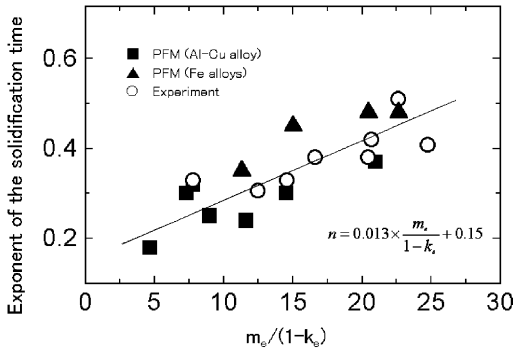


Fig. 3. Exponent of the local solidification time vs. $m_e/(1-k_e)$.

To investigate why the exponents of the local solidification time are different for each alloy, physical properties are changed artificially. The secondary arm spacing is evaluated with different values of solute diffusivities in solid and liquid, interface energy, partition coefficient and slope of liquidus in Al–1.96 mol%Cu alloy. Solute diffusivity in liquid and interface energy changes the arm spacing but does not affect the exponents. The liquidus slope significantly changes the exponent. The partition coefficient also changes both the arm spacing and the exponent. The relationship between the exponent and liquidus slope and partition coefficient is obtained by the correlation analysis of the phase-field calculation results. The exponent is proportional to a parameter $m_e/(1-k_e)$ as shown in Fig. 3. In the figure, the exponents estimated by the phase-field calculations and the experimental data are shown by the filled marks

and the open circles [30], respectively, and both are in good agreement. It is difficult to explain the change in the exponent quantitatively but the equation is useful for the estimation of the exponent for a certain alloy.

4.3. Microstructure evolution during rapid solidification

Fig. 4 shows the growth process of the dendrite during the initial stage of rapid solidification. From the initial triangle solid, the two primary arms begin to grow; one grows toward inside and the other along the surface (Fig. 4(a)). The growth rate of the latter is larger than that of the former because the melt surface is continuously cooled. Behind the tips of both arms, secondary arms grow but do not develop, as well as in the isothermal dendrite in Fig. 1. The secondary arms of the dendrite grown along the surface begin to grow competitively before the primary tip arrives at the end of the area and some of them keep growing (Fig. 4(b)). The secondary arms look like cells after the competitive growth, which are often observed in the experiments. The main difference of the competitive growth of secondary arms from the coarsening in Fig. 2 is that no melting back of the arms is observed (Fig. 4(c)). The tip of the dendrite grown initially toward inside still precedes the cells though the difference becomes small (Fig. 4(d)). The selection of growing cells continues until one of them survives finally (Fig. 4(e)). At the late stage of the calculation, the side arms begin to grow behind the tips of the cell and the transition from cell to dendrite occurs (Fig. 4(f)). The microstructure change from the cells near the surface to the dendrites observed in the rapidly solidified samples is reproduced well from the simulated results shown in the figures above.

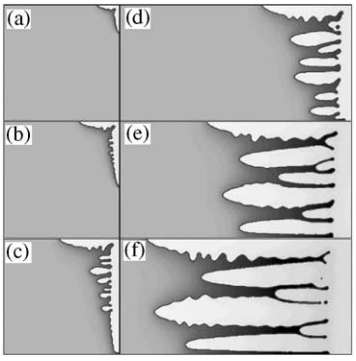


Fig. 4. Microstructure evolution during the early stage of rapid solidification: (a) 0.5, (b) 0.8, (c) 1.5, (d) 3.2, (e) 6.9 and (f) 11.0 ms.

5. Conclusion

The phase-field model for a dilute binary alloy is applied to the solidification problems of binary and ternary alloys. The governing equations are written into simplified forms with a dilute solution approximation and the relationship between the phase-field mobility and the kinetic coefficient is obtained at a thin interface limit. The model is free from the restrictions on the kinetic coefficient and the mesh size in calculation, and has a great advantage to applications. The numerical examples for the growth of isothermal free dendrite, dendrite secondary arm and dendrite in rapid solidification are demonstrated. In all cases, the characteristics of dendrite growth are reproduced and quantitatively predicted. Through these simulations, the wide applicability of the phase-field model to the solidification problems is confirmed and the range of the phase-field simulation will be further extended through the future works.

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